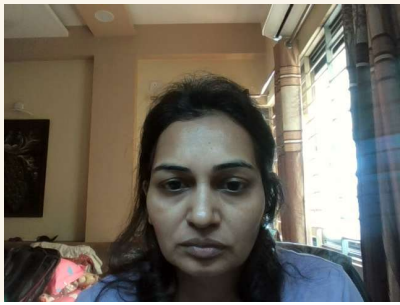
Credit Card Fraud Detection using Machine Learning

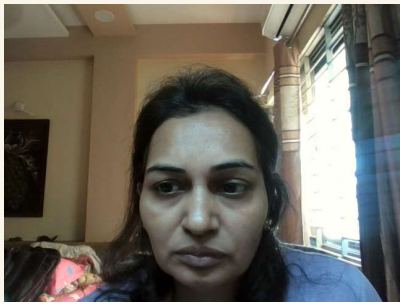
Deepika Mehta

Executive Diploma in AI & Data Science



Business Problem

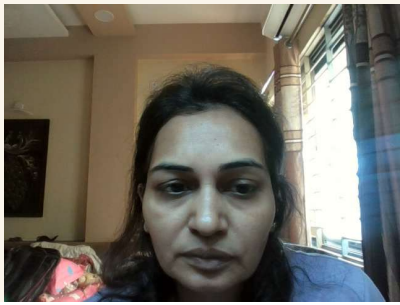
- Credit card fraud causes significant financial losses globally
- Fraud rate is very low (~0.5%) but financially impactful
- Traditional rule-based systems produce high false positives
- Need intelligent, data-driven fraud detection



Dataset Overview

- 370,000+ transactions
- 1,930 fraud cases (~0.52%)
- Highly imbalanced dataset
- Features include:
- Transaction amount
- Merchant category
- Geographic location
- Customer demographics

Key Challenge: Extreme class imbalance



Data Preparation & Feature Engineering

01

Removed irrelevant identifiers

(trans_num, cc_num)

02

Converted categorical variables

using encoding

03

Scaled numerical variables

04

Handled class imbalance

using SMOTE

05

Split into train & test sets

Objective: Build production-ready ML pipeline

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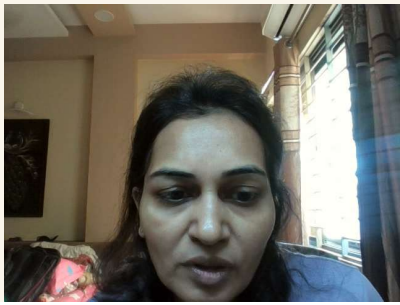
Modeling Approach

Models Used:

- *Random Forest Classifier*
- *Hyperparameter Tuning (RandomizedSearchCV)*
- *ROC-AUC optimization*
- *Threshold tuning for business balance*

Pipeline Structure:

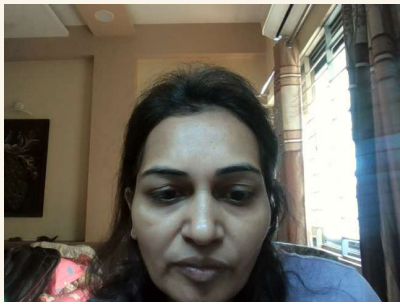
Preprocessing → SMOTE → Model



Model Performance (Before Threshold Tuning)

- *ROC-AUC: 0.95*
- *High Recall for fraud*
- *Low Precision (many false alerts)*

Problem: Model was catching fraud but flagging too many transactions.



Final Optimized Results (After Threshold Tuning)

Threshold: 0.63

Fraud Class (1):

- *Precision: 0.68*
- *Recall: 0.58*
- *F1-score: 0.62*

Fraud Detection Rate:

0.44% of transactions flagged

Balanced detection & operational feasibility

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Financial Impact Analysis ⭐

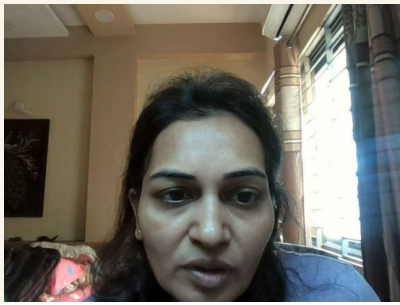
1,930	58%	1,119	$120
Test Fraud Cases	Recall	Fraud detected	Average fraud amount
		≈ cases	

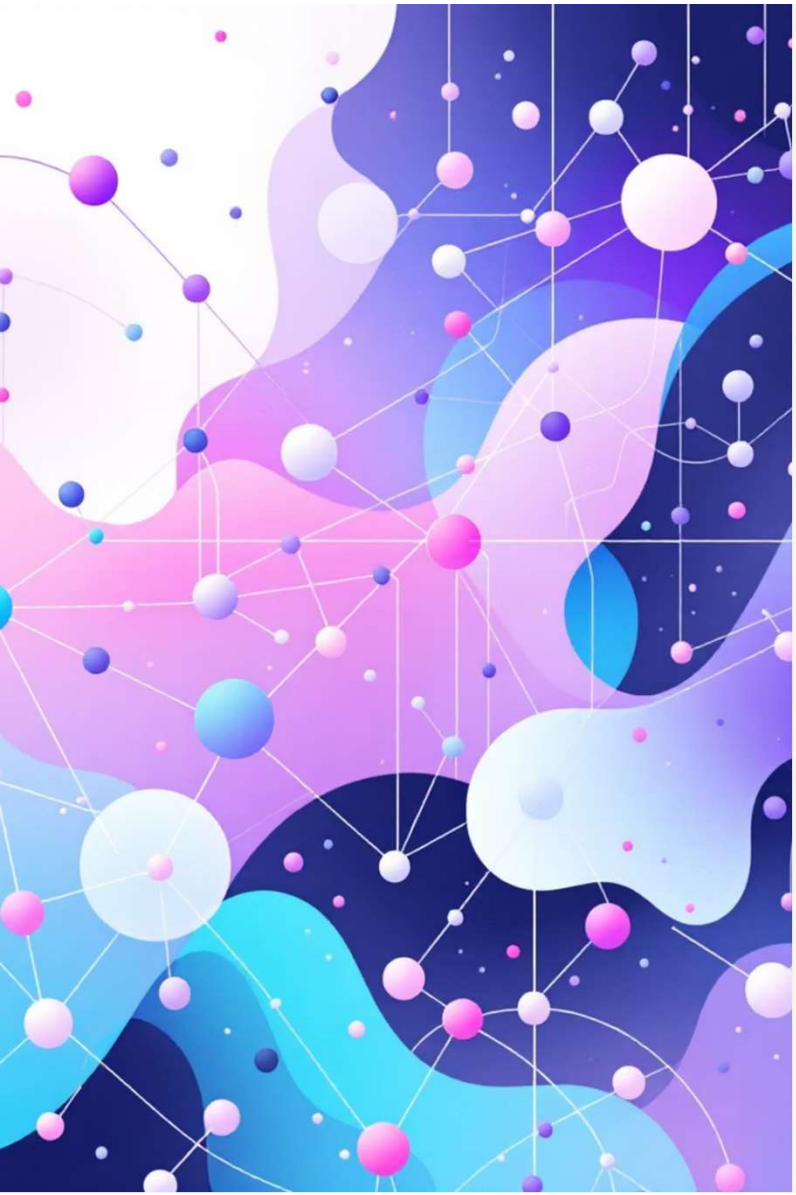
$134,280

Savings

≈ (in test sample only)

Scaled to 1M transactions: Potential annual savings in millions.





Important Variables Driving Fraud Detection

Top Predictive Features:

- *Transaction amount*
- *Merchant category*
- *Geographic distance*
- *Transaction timing patterns*
- *Customer profile attributes*

Model identifies abnormal behavior patterns.

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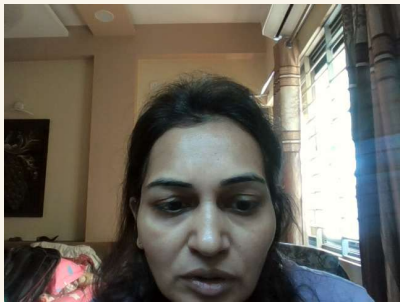
Business Recommendations

1. Deploy model with threshold = 0.63
2. Integrate with existing rule-based system
3. Retrain quarterly
4. Monitor fraud detection rate monthly
5. Use model explanations for compliance audits